

A Physiologically-Based Pharmacokinetic Model to Predict Posaconazole Suspension Exposure and Optimize Dosing Strategy in Pediatric Hematology Patients: Incorporating the Impact of Food Intake

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Objective: This study aimed to develop and validate a physiologically based pharmacokinetic (PBPK) model for posaconazole oral suspension in pediatric hematology patients to predict systemic exposure and optimize dose regimens that incorporate food intake.

Methods: A PBPK model was established and validated in adults. The model was then extrapolated to pediatric patients by integrating age-specific physiological parameters. The pediatric model was validated using published data and local therapeutic drug monitoring results. The validated model was employed to simulate posaconazole exposure across different age groups, dosing regimens, and meal conditions.

Results: The adult PBPK model showed good predictability, with fold errors within 0.5–2.0 and a geometric mean fold error < 2.0. Sensitivity analysis identified the bile salt solubilization ratio, gastric pH, and hepatic UGT1A4 as critical determinants. The pediatric model accurately predicted plasma concentrations, with a mean absolute prediction error of 12.3%. Simulations revealed that achieving prophylactic trough concentration (C_{min}) is challenging under fasted conditions, while fed administration significantly increases C_{min} . C_{min} s increased with age and meal fat/calorie content, but no significant further increase was observed across all age groups beyond 6 mg/kg dose levels under fed conditions, with this plateau pattern consistent for both three- and four-times daily regimens. The oral suspension was unreliable in children aged < 2 years. In contrast, a high-fat, high-calorie meal was necessary for children aged 2–6 years, and a moderate-fat, moderate-calorie meal for those aged 7–12 years, whereas a low-fat, low-calorie meal was sufficient for those aged ≥ 13 years.

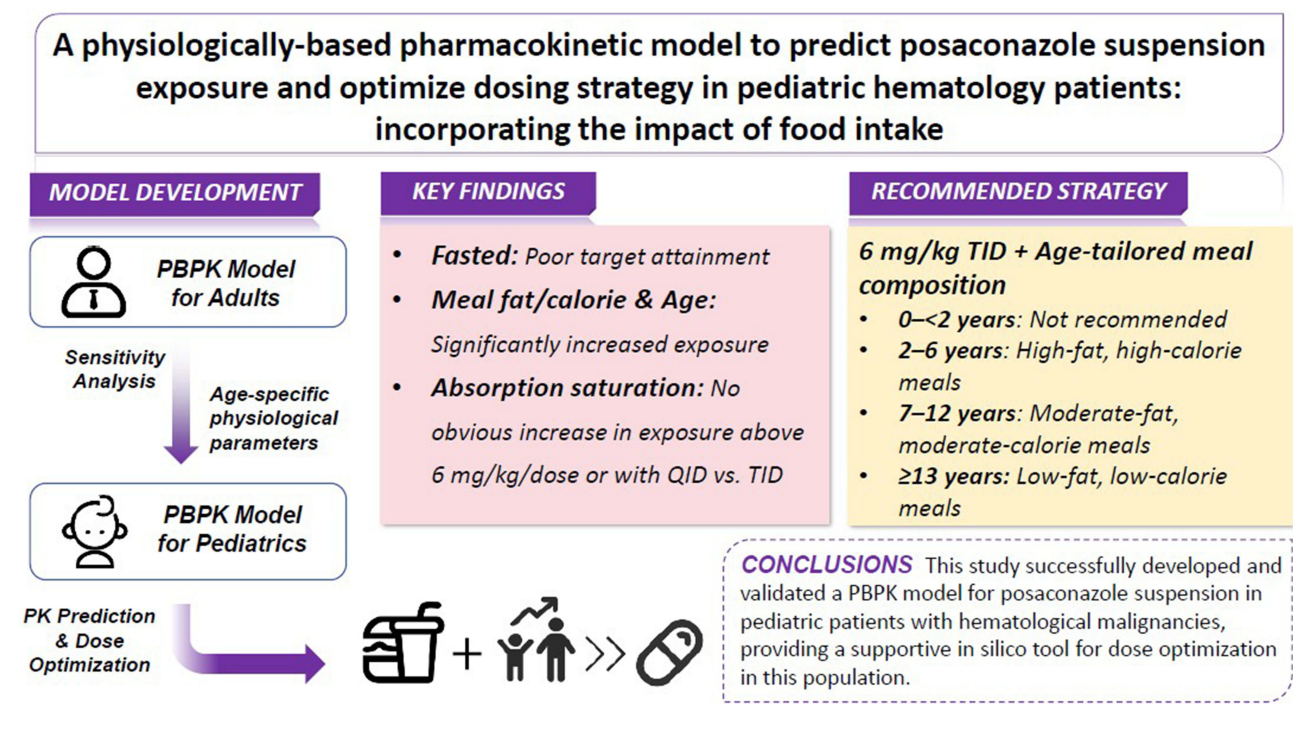
Conclusion: This study successfully developed and validated a PBPK model for posaconazole suspension in pediatric hematology patients, which could serve as a supportive in silico tool for model-informed dose optimization in this population. Further prospective clinical validation is still required to support its real-world dosing decision application.

Keywords: posaconazole oral suspension, physiologically based pharmacokinetic model, pediatrics, dose optimization, food intake

Introduction

Posaconazole is a broad-spectrum triazole antifungal agent with activity against common pathogens such as *Candida* and *Aspergillus*, as well as emerging resistant fungi, including *Cryptococcus neoformans*, *Fusarium*, and *Zygomycetes*.^{1,2} Posaconazole suspension is primarily indicated for prophylaxis of invasive fungal infections (IFIs) in patients aged 13 years and older who are at high risk of immunocompromise, including those with hematologic malignancies or post-hematopoietic stem cell transplantation (HSCT) graft-versus-host disease (GVHD).³ However, in clinical practice, posaconazole suspension is often used off-label in children under 13 years of age due to its liquid formulation.⁴

Graphical Abstract



Compared with intravenous (IV) formulation and delayed-release tablet, posaconazole suspension exhibits greater inter- and intra-individual pharmacokinetic (PK) variability, particularly in pediatric patients. This variability is primarily driven by key factors influencing its absorption, including gastric pH, food intake, and concomitant medications, such as proton pump inhibitors (PPIs).^{5,6} Furthermore, its metabolism is complicated by UDP-glucuronidation and susceptibility to P-glycoprotein (P-gp) efflux.⁷ Consequently, this heightened variability may lead to unpredictable plasma concentrations and an increased risk of subtherapeutic exposure and prophylactic failure, defined as a trough concentration ($C_{\min}$) below 700 ng/mL for prophylaxis.^{8,9}

Physiologically based pharmacokinetic (PBPK) modeling offers a promising approach for PK prediction in the pediatric population because it considers dynamic changes in physiological parameters such as organ size, composition, blood flow rates, and the expression of drug-metabolizing enzymes or transporters throughout growth and development.^{10,11} To date, the vast majority of published posaconazole PBPK studies are focused on adult subjects and drug-drug interaction (DDI) assessments, while PBPK modeling of posaconazole suspension in pediatric patients remains insufficient.^{12,13}

This study aimed to develop and validate a PBPK model for posaconazole suspension in pediatric patients with hematological malignancies. By integrating age-specific physiological parameters and the impact of food intake, the model will enable individualized dosing regimens, mitigating the risks of sub-therapeutic exposure and prophylactic failure in this vulnerable population.

Materials and Methods

PBPK Model Development Workflow

Physicochemical and biopharmaceutical property parameters of posaconazole were systematically collected for modeling, such as solubility, permeability, lipophilicity, plasma protein free fraction (F_{up}), and blood-to-plasma drug concentration ratio (R_{bp}).^{13,14} Formulation-related parameters such as dissolution curves in different media, particle size, particle

density, etc. were also retrieved and organized.¹⁵ The basic modeling parameters of posaconazole are listed in [Table S1](#). Additionally, the literature-reported plasma concentration-time (C-T) data of human subjects following IV or oral administration were compiled.^{16–21}

This study followed a structured PBPK modeling workflow, as shown in [Figure 1](#). An adult model was first developed using IV data to establish and optimize the distribution and elimination parameters of posaconazole. Subsequently, an oral PBPK model was developed using the PK data of both oral tablet and suspension to optimize the formulation-specific dissolution and absorption parameters. A parameter sensitivity analysis (PSA) was performed to identify key influencers and assess the robustness of the model. The suspension model for adults was then extrapolated to pediatric populations by integrating age-specific physiological parameters, enabling exposure prediction and dose optimization. All models were evaluated by comparing the simulated exposures with the observed clinical PK data.

Software

The PBPK model was constructed using GastroPlus® (version 9.8.3; Simulations Plus, Inc., USA). The measured C-T data points were extracted from the C-T curves using Digit (version 1.0.4, Simulations Plus, Inc., USA).

PBPK Model for Adults

Development of a PBPK model for IV administration in healthy adults was initiated. The key model parameters, including the formulation, dose, study population, and PK settings, are summarized in [Table S2](#). Published C-T data from a clinical study involving a 300 mg IV infusion (duration: 0.5 h) in healthy Chinese subjects were used to directly estimate the distribution and elimination characteristics of posaconazole,¹⁶ thereby enabling accurate characterization of

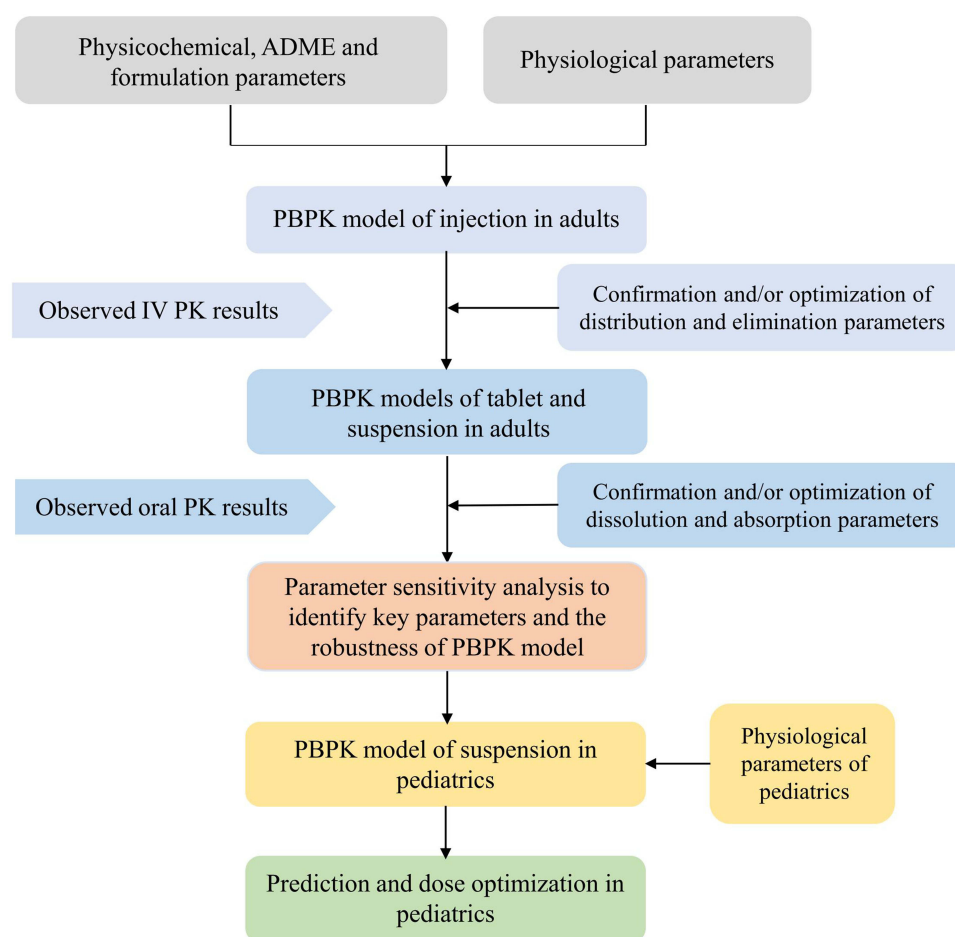


Figure 1 Workflow of PBPK modeling of posaconazole in pediatrics.

its clearance and volume of distribution. To further evaluate the model, simulations were performed for two additional dosing scenarios: a 250 mg IV infusion and a 400 mg IV bolus.^{17,18} Relevant demographic variables and administration details were adjusted in the model accordingly, while all other drug-specific parameters remained consistent with the initial model. The predictive performance of the model was assessed by comparing the simulated C-T profiles with the corresponding observed PK data for each dosing regimen.

Following the IV model, an oral PBPK model was developed for delayed-release tablet and suspension in adults. The distribution and elimination parameters were consistent with those established in the prior IV model. Demographic information from the respective clinical studies and formulation-specific parameters were integrated (Table S2).^{19–21} Parameters governing drug dissolution and intestinal absorption were incorporated, and the oral absorption model was coupled to simulate C-T profiles under different meal conditions (fasted and fed). This framework enabled the optimization and validation of the tablet model. Subsequently, employing the same methodology, a suspension model was developed, optimized, and validated across varying doses and meal conditions to evaluate its predictive capability.

Sensitivity Analysis of Model Parameters

Based on the established PBPK model for healthy subjects receiving a 200 mg oral suspension of posaconazole under both fasted and fed conditions, a PSA was conducted to investigate the key factors influencing systemic exposure to posaconazole in vivo. The factors and their corresponding ranges are presented in Table S3.

PBPK Model for Pediatrics

The pediatric PBPK model for posaconazole suspensions was developed by adapting a previous adult model, incorporating pediatric physiological parameters retrieved from the built-in Pediatric Physiology (PEAR) module of the software, such as age-scaled organ weights/volumes, tissue blood perfusion rates, metabolic enzyme and transporter expression, and pre-/postprandial intestinal bile salt concentrations. Population simulations were performed in addition to single simulations, considering the potentially substantial interindividual variability among pediatric patients. Details regarding the demographic information, dosing regimens, and parameters of the virtual cohort are provided in Table S4. The accuracy of the model was verified against both clinical PK data reported in the literature and therapeutic drug monitoring (TDM)-derived C_{min} s from our center (Table S5).^{22,23}

Model Verification

To quantitatively evaluate the accuracy of the model, multiple metrics were systematically employed.²⁴ The accuracy of the exposure-level predictions was assessed by calculating the fold error (FE) and geometric mean fold error (GMFE) for key PK parameters, such as C_{max} , T_{max} and area under the plasma concentration-time curve (AUC_{0-t} , AUC_{0-inf}), as defined by Equations 1 and 2. If C_{max} and AUC data are unavailable, the prediction error (PE) and mean absolute prediction error (MAPE) are derived by comparing the predicted and literature-reported mean plasma drug concentrations, as defined by Equations 3 and 4. The validation criteria required FE within 0.5–2, $GMFE < 2.0$, and $MAPE \leq 30\%$.

$$FE = \frac{\text{Simulated}}{\text{Observed}} \quad (1)$$

$$GMFE = 10^{\frac{1}{n} \sum \left| \lg \left(\frac{\text{Simulated}}{\text{Observed}} \right) \right|} \quad (2)$$

$$PE(\%) = \frac{\text{Simulated} - \text{Observed}}{\text{Observed}} \times 100\% \quad (3)$$

$$MAPE(\%) = \frac{1}{n} \sum |PE| \quad (4)$$

Prediction of Posaconazole PK and Dose Strategies in Pediatric Patients

Based on the validated PBPK model, we modified the age-specific physiological parameters to reflect the characteristics of pediatric patients. We then simulated *in vivo* exposure to posaconazole suspension following oral administration under various conditions, including different age groups, dose levels, fasted and fed (moderate-fat, moderate-calorie) conditions, dosing frequencies, and meal contents. Age groups were categorized as 0–< 2, 2–6, 7–12, and 13–18 years, with dose levels of 6, 12, and 18 mg/kg, respectively. Each dose was administered three times daily (TID) or four times daily (QID). Additionally, the impact of different meal conditions was simulated at a dose of 6 mg/kg TID, with the caloric content and fat intake set to the software's default values (Table S6). Population simulations were conducted with a male-to-female ratio of 50%, comprising 500 individuals in total. All other parameters were set using the default software. Each simulation was run for 168 h, with the concentration at the 168th hour defined as the steady-state $C_{\min}$.

Results

PBPK Model for Adults

PBPK Model for IV Administration of Posaconazole in Adults

The mean plasma C-T profiles following IV administration of different posaconazole doses in healthy adults, comprising both simulated and observed data, are shown in Figure 2. As shown in Table 1, for the various IV dosing regimens in healthy adults, the FE values ranged from 0.9 to 2.1, and the GMFE values ranged from 1.1 to 1.8.

PBPK Model for Oral Tablet and Suspension of Posaconazole in Adults

The mean plasma C-T profiles following a single oral administration of posaconazole tablet and suspension at varying doses under different meal conditions in healthy adults, incorporating both simulated and observed data, are presented in Figures 3 and 4. As shown in Table 2, for the various oral dosing regimens in healthy adults, the FE values ranged from 0.5 to 1.5, and the GMFE values ranged from 1.1 to 1.3.

Sensitivity Analysis of Model Parameters

The impact of various parameters on exposure ($C_{\max}$, $AUC_{0-\infty}$) following 200 mg oral administration of posaconazole suspension in healthy subjects under fasted and fed conditions is illustrated in Figure 5. The results indicate a significant positive correlation between *in vivo* posaconazole exposure and the bile salt solubilization ratio under fasted conditions. In contrast, under fed conditions, drug exposure initially increased significantly with increasing bile salt solubilization ratio until reaching 1.4×10^5 , beyond which no further significant increase was observed. Gastric pH and UGT1A4 enzyme expression, especially hepatic expression, significantly influenced posaconazole exposure, implying that its variability in pediatric patients may contribute to substantial inter-individual PK differences. In addition, systemic exposure to posaconazole was inversely correlated with R_{bp} and F_{up} , and positively correlated with F_{ent} .

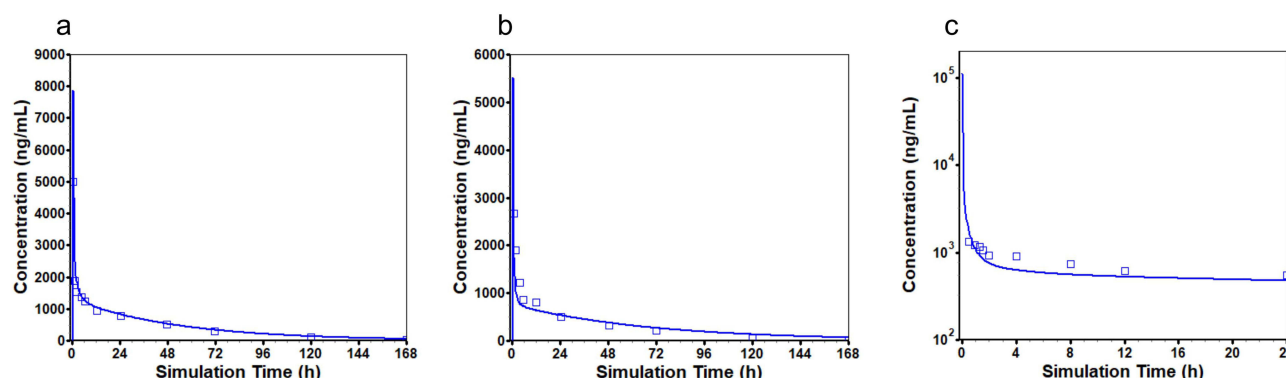


Figure 2 Average plasma C-T curves following single administration of posaconazole in healthy adults. (a) 300 mg IV infusion, (b) 250 mg IV infusion, (c) 400mg IV bolus. Squares represent observed values and solid lines represent simulated values.

Table 1 Simulated and Observed PK Parameters of IV Posaconazole in Healthy Adults

Dose	Parameters	C _{max} (ng/mL)	AUC _{0-t} (ng h/mL)	AUC _{0-inf} (ng h/mL)
300mg infusion	Simulated	7859.5	71,760	74,650
	Observed	5000.0	65,990	67,600
	FE	1.6	1.1	1.1
250mg infusion	Simulated	5519.3	50,750	55,330
	Observed	2660.0	46,370	47,130
	FE	2.1	1.1	1.2
400mg bolus	Simulated	NA	16120	76,490
	Observed	NA	17170	80,820
	FE	NA	0.9	0.9
Adults IV GMFE		1.8	1.1	1.1

Abbreviation: NA, not applicable.

PBPK Model for Pediatrics

The PBPK model for oral posaconazole suspension across different doses in pediatric patients with hematologic malignancies was successfully established and validated using clinical PK data, as illustrated in [Figure 6](#). In individual simulations, the MAPE between the simulated and measured plasma concentrations of posaconazole oral suspension across different dosing regimens in pediatric patients was 12.3%. The detailed simulated and observed plasma concentrations, along with corresponding PE values, are presented in [Table S5](#). In population simulations, the simulated PK profiles and 90% confidence intervals (CIs) for the virtual population effectively encompassed the observed mean C-T data. Furthermore, the upper and lower bounds of the individual 90% CIs largely covered the variability observed in clinical data.

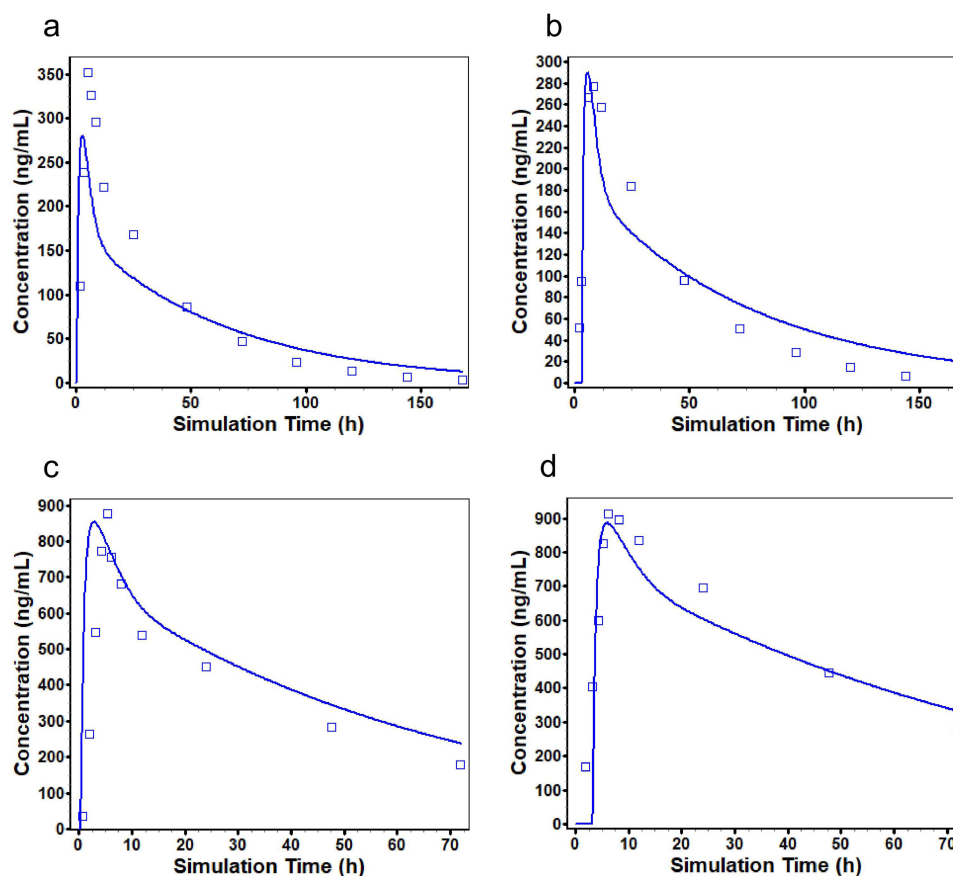


Figure 3 Average plasma C-T curves following single administration of posaconazole tablet in healthy adults. (a) 100 mg, fasted, (b) 100 mg, fed, (c) 300 mg, fasted, (d) 300 mg, fed. Squares represent observed values and solid lines represent simulated values.

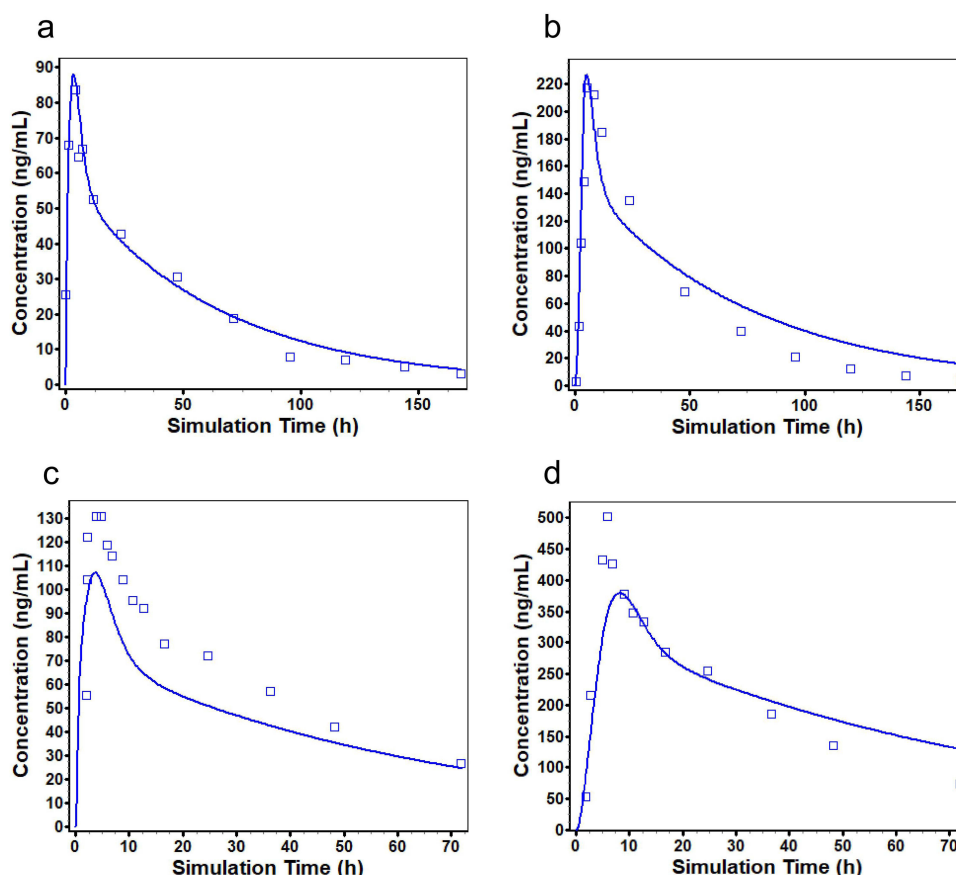


Figure 4 Average plasma C-T curves following single administration of posaconazole suspension in healthy adults. (a) 100 mg, fasted, (b) 100 mg, fed, (c) 200 mg, fasted, (d) 200mg, fed. Squares represent observed values and solid lines represent simulated values.

Prediction of Posaconazole PK and Dose Strategies in Pediatric Patients

Prediction of Posaconazole PK Under Fasted Condition

The population PK simulation results of the posaconazole oral suspension at doses of 6, 12, and 18 mg/kg TID for virtual pediatric patients under fasted conditions are presented in [Figure 7](#). Under fasted conditions, the mean C_{min} s of posaconazole oral suspension increased with both dose and age in pediatric patients. However, achieving the target

Table 2 Simulated and Observed PK Parameters of Oral Posaconazole in Healthy Adults

Dose	Meal Condition	Parameters	C_{max} (ng/mL)	T_{max} (h)	AUC_{0-t} (ng h/mL)	AUC_{0-inf} (ng h/mL)
100mg tablet	Fasted	Simulated	280.4	2.7	10,930	11,740
		Observed	352.0	5.2	11,620	11,800
		FE	0.8	0.5	0.9	1.0
	Fed	Simulated	290.2	5.7	12,910	14,340
		Observed	227.0	8.1	11,930	12,050
		FE	1.3	0.7	1.1	1.2
300mg tablet	Fasted	Simulated	853.8	2.9	31,710	47,220
		Observed	877.0	5.4	26,920	36,580
		FE	1.0	0.5	1.2	1.3
	Fed	Simulated	886.2	6.0	36,740	63,100
		Observed	914.0	6.1	39,290	54,620
		FE	1.0	1.0	0.9	1.2

(Continued)

Table 2 (Continued).

Dose	Meal Condition	Parameters	C _{max} (ng/mL)	T _{max} (h)	AUC _{0-t} (ng h/mL)	AUC _{0-inf} (ng h/mL)
100mg suspension	Fasted	Simulated	88.3	3.5	3653.7	3926.1
		Observed	83.6	4.3	3534.8	3689.7
		FE	1.1	0.8	1.0	1.1
	Fed	Simulated	226.8	5.1	10,460	11,600
		Observed	217.0	5.1	8986.7	9163.1
		FE	1.0	1.0	1.2	1.3
200mg suspension	Fasted	Simulated	105.7	3.7	3407.4	5036.8
		Observed	131.0	3.8	4317.3	5738.6
		FE	0.8	1.0	0.8	0.9
	Fed	Simulated	378.8	8.2	15,220	25,100
		Observed	502.0	5.8	14,320	17,280
		FE	0.8	1.4	1.1	1.5
Adults oral GMFE			1.2	1.3	1.1	1.2

prophylactic trough level of 700 ng/mL remains challenging with this formulation under fasted conditions, especially in patients aged 12 years and younger.

Prediction of Posaconazole PK Under Fed Condition

Figure 8 shows the population PK simulation results of the posaconazole oral suspension at doses of 6, 12, and 18 mg/kg administered TID and QID in virtual pediatric patients under moderate-fat, moderate-calorie meals. Compared with fasted, fed administration significantly increased systemic exposure to the posaconazole suspension. C_{min} increased with age, but plateaued above 6 mg/kg, with similar results for QID and TID. The target C_{min} attainment rates were 58.2% for age < 2 years, 67.8% for age 2–6 years, 75.8% for age 7–12 years, and 90.8% for age ≥ 13 years at a dose of 6 mg/kg TID. However, approximately 22.0% of children over 13 years of age receiving weight-based dosing may have C_{mins} exceeding the well-tolerated upper limit of 3750 ng/mL.

Prediction of Posaconazole PK Under Different Fat and Calorie Meal Conditions

Figure 9 illustrates the impact of dietary fat and calorie content on exposure to posaconazole oral suspension. These results indicate that increasing both the fat and calorie content of a meal significantly elevates the C_{mins} of posaconazole. For children aged < 2 years, even with a high-fat, high-calorie meal, the target C_{min} attainment rate was only 67.4%. In contrast, for children aged 2–6 years, this rate reached 91.6% under a high-fat, high-calorie meal, while for children aged 7–12 years, the attainment rate was 75.8% under a moderate-fat, moderate-calorie meal. In children aged 13 years and older, a low-fat, low-calorie meal was sufficient for 86.2% of the patients to attain the target C_{min}.

Discussion

As a Biopharmaceutics Classification System (BCS) class II compound, which is characterized by low solubility but high permeability, the absorption of posaconazole oral suspension leads to significant intra- and inter-individual PK variability.²⁵ Studies in healthy adult volunteers have shown that after a single 200 mg dose of posaconazole oral suspension is administered with a non-high-fat meal or a high-fat meal, the average AUC and C_{max} are approximately threefold and fourfold higher, respectively, compared to administration under fasted condition.³ Conversely, co-administered with drugs that inhibit gastric acid secretion, such as proton pump inhibitors (PPIs) and histamine (H₂) receptor antagonists, as well as drugs that alter gastrointestinal motility, such as metoclopramide, has been associated with a significant reduction in posaconazole bioavailability.^{4,6} Although a large body of PK studies have focused on adult patients or population pharmacokinetic (PPK) models,^{10–14} they have generally failed to account for the dynamic changes in physiological parameters that occur during pediatric growth and development. Furthermore, the inherent physiological limitations of gastrointestinal absorption have not yet been fully addressed.¹⁵

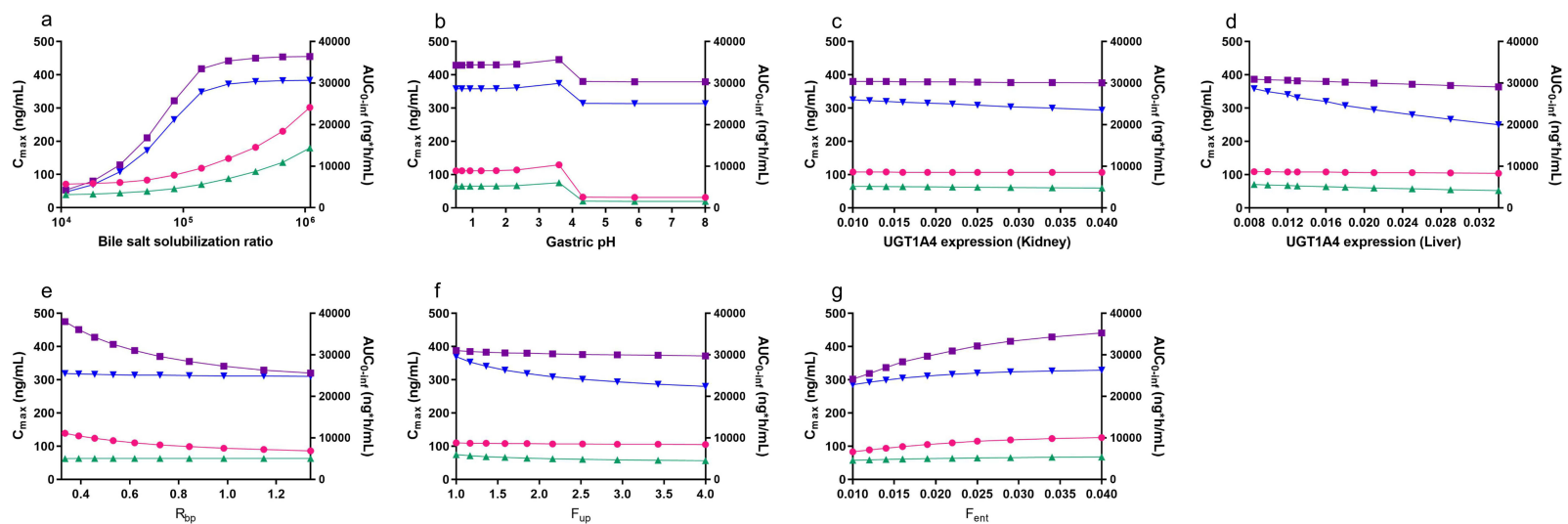


Figure 5 Impact of basic model parameters on systemic exposure of posaconazole suspension. (a) bile salt solubilization ratio, (b) gastric pH, (c) UGT1A4 expression (kidney), (d) UGT1A4 expression (liver), (e) R_{bp} , (f) F_{up} , (g) F_{ent} . Pink circles represent C_{max} s under the fasted condition, purple squares represent C_{max} s under the fed condition, green triangles represent $AUC_{0-\infty}$ s under the fasted condition and blue triangles represent $AUC_{0-\infty}$ s under the fed condition.

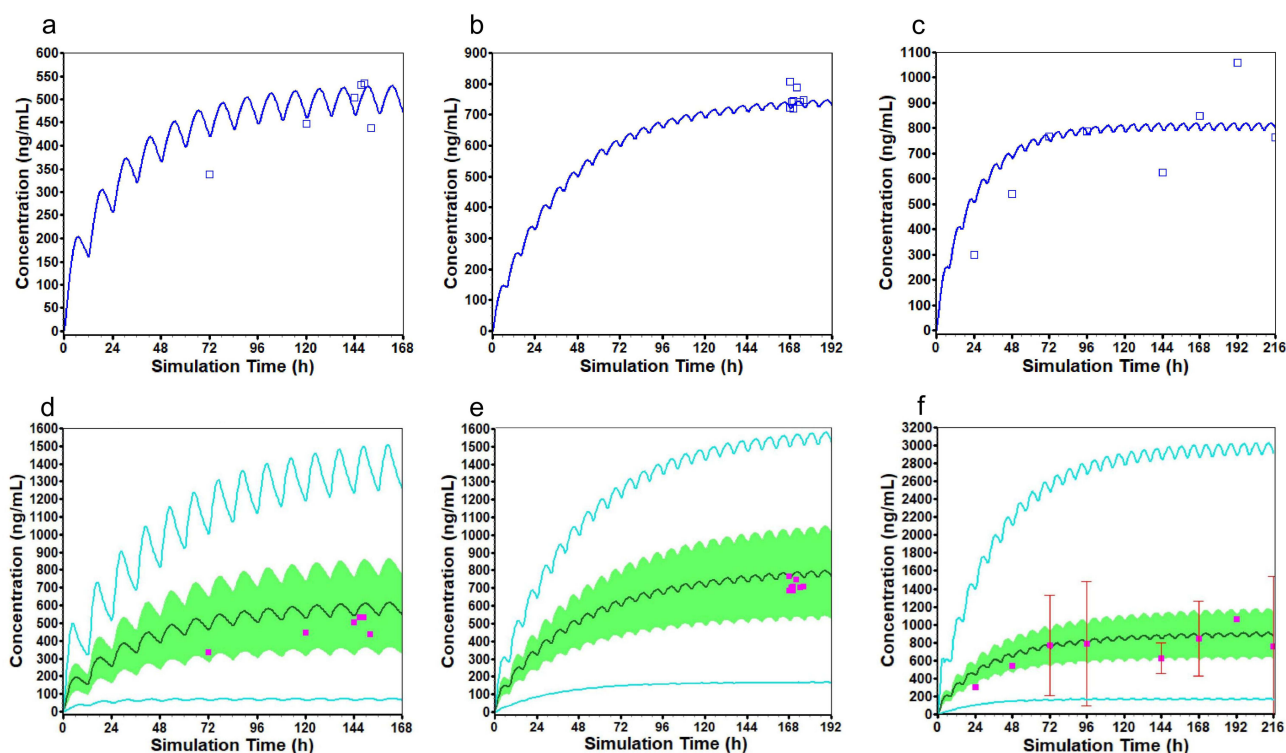


Figure 6 Average plasma C-T curves following oral administration of posaconazole suspension in pediatric patients. (a) 9 mg/kg, Bid, (b) 120mg/m², TID, (c) 5.3mg/kg, TID, (d) 9 mg/kg, BID, population simulation, (e) 120mg/m², TID, population simulation, (f) 5.3mg/kg, TID, population simulation. In single simulations, squares represent observed values and solid lines represent simulated values. In population simulations, red squares represent observed values, with red error bars indicating the corresponding standard deviations. Black lines represent simulated values. Blue lines represent the 90% CIs for individual predictions, and green areas represent the 90% CIs for the mean population predictions.

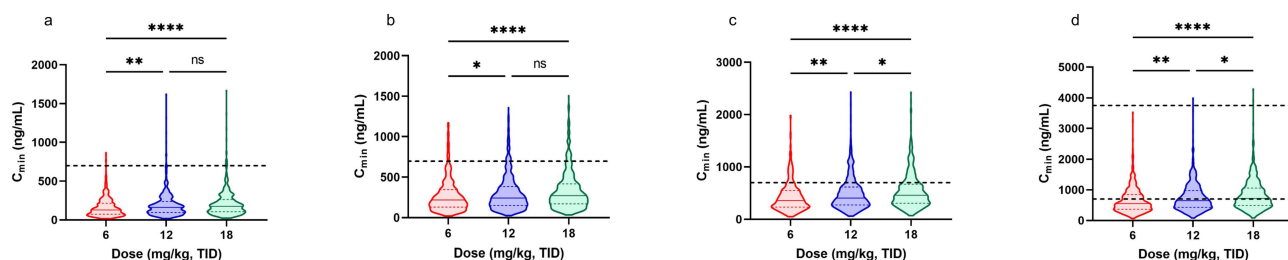


Figure 7 Population simulations of posaconazole C_{min} s for different age groups following various TID dose levels under fasted conditions. (a) 0 – 2 years, (b) 2–6 years, (c) 7–12 years, (d) 13–18 years. Black dashed lines represent the lower and upper target C_{min} s (700 and 3750 ng/mL, respectively). Three horizontal lines inside each violin plot denote the first quartile (25th percentile), the median (50th percentile) and the third quartile (75th percentile), respectively. * P value < 0.05, ** P value < 0.01, **** P value < 0.0001, ns P value $\geq$ 0.05.

This study successfully developed and validated PBPK models for posaconazole in healthy adults using different doses, formulations, and meal conditions. The majority of FE and GMFE values were within the acceptable ranges for both IV and oral dosing regimens in healthy adults. The FE value 2.1 for C_{max} following 250 mg infusion was primarily attributable to the measurement error of C_{max} resulting from the timing of blood sampling. The shape-related deviation was observed for the 200 mg fasted suspension, likely reflecting inter-individual variability in fasting gastrointestinal physiology and peripheral distribution volume. These models reasonably and accurately reflect the in vivo absorption, distribution, metabolism, and excretion (ADME) process of posaconazole in adults.

The bile salt solubilization factor was identified as a key parameter influencing posaconazole exposure, primarily because posaconazole is a BCS class II compound, for which dissolution is the rate-limiting step in absorption. This factor quantifies bile-mediated micellar solubilization, which not only governs the absorbable drug concentration in the

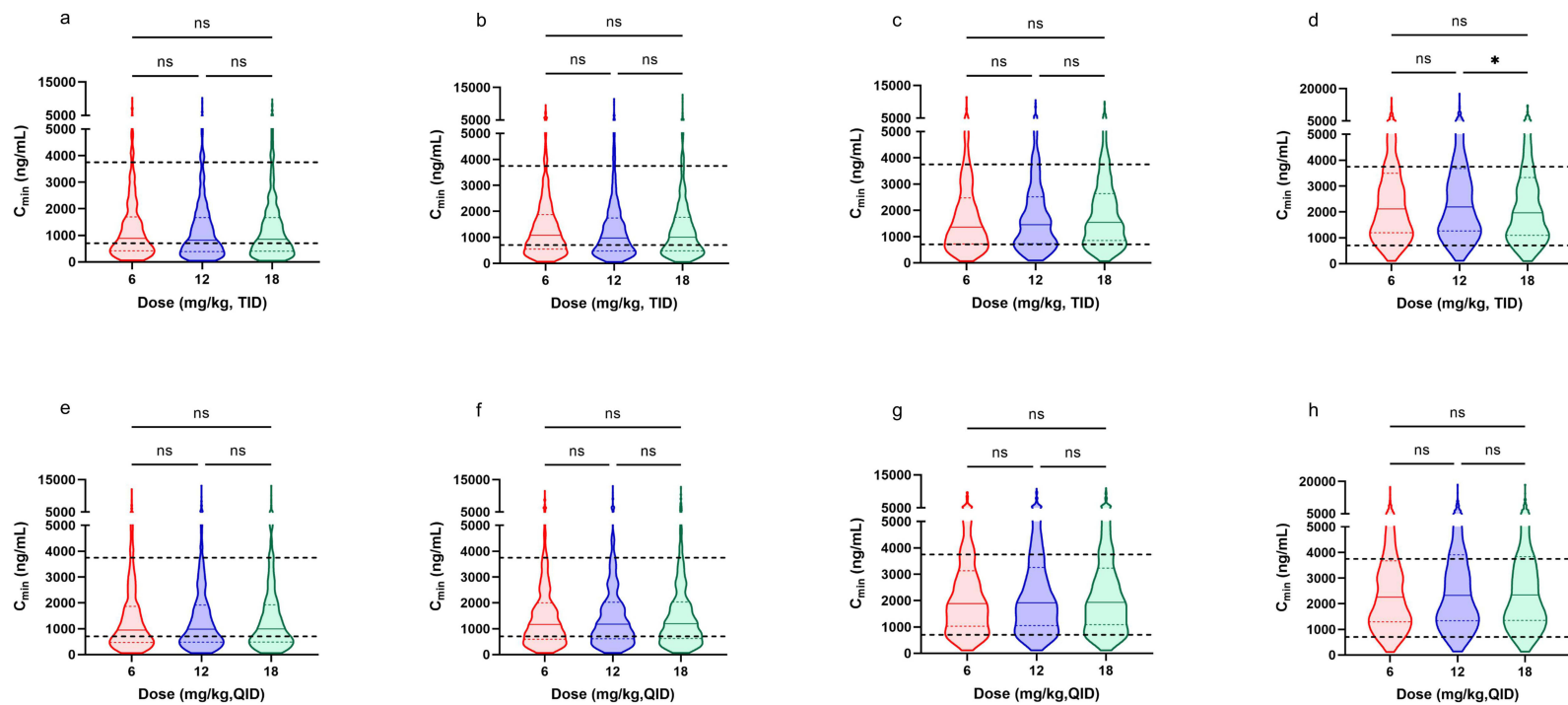


Figure 8 Population simulations of posaconazole C_{min} s for different age groups following various TID and QID dose levels under moderate-fat, moderate-calorie meal conditions. (a) 0 – < 2 years, TID, (b) 2–6 years, TID, (c) 7–12 years, TID, (d) 13–18 years, TID, (e) 0 – < 2 years, QID, (f) 2–6 years, QID, (g) 7–12 years, QID, (h) 13–18 years, QID. Black dashed lines represent the lower and upper target C_{min} s (700 and 3750 ng/mL, respectively). Three horizontal lines inside each violin plot denote the first quartile (25th percentile), the median (50th percentile) and the third quartile (75th percentile), respectively. * P value < 0.05, ns P value $\geq$ 0.05.

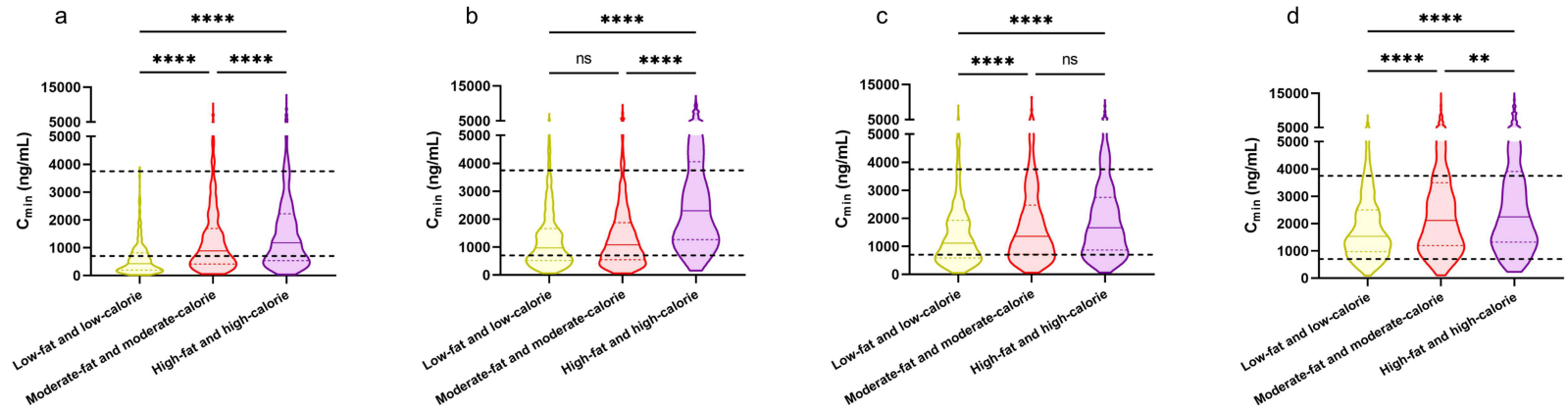


Figure 9 Population simulations of posaconazole C_{min} s under different fat and calorie meal conditions. (a) 0 – < 2 years, (b) 2–6 years, (c) 7–12 years, (d) 13–18 years. Black dashed lines represent the lower and upper target C_{min} s (700 and 3750 ng/mL, respectively). Three horizontal lines inside each violin plot denote the first quartile (25th percentile), the median (50th percentile) and the third quartile (75th percentile), respectively. ** P value < 0.01, **** P value < 0.0001, ns P value $\geq$ 0.05.

intestinal lumen, but also elucidates the physiological basis for the marked food effect, particularly dietary fat, on drug exposure.^{26,27} As a weakly basic drug, posaconazole exhibits pH-dependent solubility. The baseline gastric pH is reported to be approximately 1.3 under fasted conditions and 4.9 under fed conditions. After PPI administration, gastric pH rises variably to 2.8–6.1 depending on PPI type, dose and regimen.²⁸ From the PSA results of this study, it can be inferred that fluctuations in gastric pH may produce an observable impact on the dissolution and subsequent absorption of posaconazole under fasted conditions. This has been confirmed in several studies.²⁹ However, in the widely adopted fed conditions in clinical practice, the additional gastric pH elevation mediated by PPIs may not cause clinically significant changes to the exposure of posaconazole. UGT1A4-mediated glucuronidation is considered the primary metabolic pathway of posaconazole,³⁰ and variability in the expression of this enzyme may directly influence its metabolic clearance. As a strong inhibitor of CYP3A4, posaconazole mainly acts as a perpetrator drug in DDIs, altering the in vivo exposure of other CYP3A4 substrate medications. There is rarely DDIs between posaconazole and most concurrently administered chemotherapeutic agents that can significantly modify the PK profile of posaconazole. Changes in R_{bp} and F_{up} may alter free drug concentrations, which could subsequently affect the rates of drug metabolism and clearance and thus potentially influence drug exposure. F_{ent} is thought to represent the efficiency of drug transport across intestinal epithelial cells into the portal circulation.³¹ An increase in F_{ent} would facilitate the passage of free drugs across the basolateral membrane of enterocytes into the portal blood, consequently increasing systemic exposure. Taken together, owing to differences in physiological parameters, especially in the pediatric population, each of the above factors could contribute to the PK variability of posaconazole.

A pediatric PBPK model was successfully developed and validated by incorporating age-related physiological parameters into an adult model. Virtual populations were generated using Monte Carlo simulations to compare the estimated inter-individual variability with clinical observations. The wide 90% CIs presented in this population simulation reflect the well-documented high inter-individual pharmacokinetic variability of posaconazole oral suspension in pediatric populations. This alignment suggests that the parameters and their incorporated variances sufficiently reflect clinical PK diversity, justifying the application of the model for further pediatric PK predictions.

Pediatric model-based simulations demonstrated that postprandial administration of posaconazole oral suspension significantly enhanced systemic exposure compared to fasted. Notably, this increase was positively correlated with the fat and caloric contents of the meal. This finding aligns well with the drug label recommendation, which specifies that posaconazole should be administered with or immediately following a meal.³ The underlying mechanism primarily involves postprandial bile salt secretion to form mixed micelles that solubilize posaconazole, thereby enhancing its intestinal permeability and enterocyte uptake.³² The regulatory effect of dietary fat content on this process is mediated through the cholecystokinin (CCK) pathway. In the fasted condition, 70% of bile salts secreted by the liver are stored in the gallbladder, maintaining a low baseline bile salt level in the intestinal lumen. Postprandially, fatty acids in the duodenum stimulate the intestinal mucosa to secrete CCK. A high-fat diet can induce maximal CCK release to drive complete gallbladder emptying and high-concentration bile salt efflux into the gut.³³ However, there are marked physiological differences in bile salt secretion capacity between children and adults. Postprandial bile salt concentrations increase from approximately 0.01–0.06 mmol/L to 4–7.5 mmol/L in children, compared with an increase from 4–6 mmol/L to 10–20 mmol/L in adults.^{34–36} This difference implies that children are more dependent on meal-induced solubilization, yet achieve lower solubilization efficiency. This mechanism is further supported by several pediatric PK studies underscoring the critical role of meal composition in modulating systemic exposure.^{6,37}

Our simulations further revealed that posaconazole absorption and exposure became more reliable with increasing age in children. The fundamental reason for this age-dependent difference may be the limitation of bile salt secretion in younger children. The 67.4% target attainment rate for children under 2 years under a high-fat, high-calorie meal is still clinically unacceptable for antifungal prophylaxis, as more than 30% of this vulnerable population will face subtherapeutic exposure and elevated infection risk. Similarly, the 67.8% rate under a moderate-fat, moderate-calorie meal in the 2–6 years group fails to meet the reliable prophylaxis threshold, indicating that a high-fat, high-calorie meal is necessary in this age group. A previous PBPK study supports the conclusion that due to solubility-limited absorption, the use of oral suspension is impractical and inevitably leads to subtherapeutic exposure in the majority of children under 7 years of age.³⁸ This finding explains why current prescribing information and clinical guidelines

emphasize the need for TDM in pediatric patients and recommend the preferential use of delayed-release formulations or IV formulations in children aged ≤ 12 years. Unlike the oral suspension, IV posaconazole provides predictable systemic exposure independent of gastrointestinal status, a critical advantage in pediatric patients who may be unable to receive oral medications with food due to mucositis, nausea, or peri-procedural fasting. It was indicated that approximately 22.0% of adolescents over 13 years would have $C_{\min}$ values above 3750 ng/mL under weight-based dosing. However, as this age group conventionally receives the same fixed dose of 200 mg TID as adults in clinical practice,³ this figure should not be directly extrapolated to real-world settings. From a risk-benefit perspective, the current fixed-dose approach appears to be the more prudent choice. For patients with $C_{\min}$ s consistently exceeding 3750 ng/mL, TDM-guided dose reduction may be considered.

Notably, our study found that posaconazole exposure plateaued at doses above 6 mg/kg under fed conditions, with no significant increase in exposure upon further dose escalation. This phenomenon is likely attributable to the saturable nature of the bile salt-mediated solubilization. Clinically, this means that further dose escalation provides no additional systemic exposure while potentially leading to gastrointestinal adverse effects such as nausea, vomiting, abdominal pain, and diarrhea,^{22,39,40} as excess undissolved drug may cause local irritation or osmotic effects in the intestinal lumen. Therefore, the 6 mg/kg dose represents a practical upper limit for this formulation. Interestingly, our results showed no significant difference in systemic exposure between TID and QID dosing regimens under fed conditions. This may be consistent with our finding of a dose-exposure plateau above 6 mg/kg, reflecting the same solubility-limited mechanism. Once the administered dose saturates the bile salt-mediated solubilization, excess drug remains unabsorbed. When both TID and QID regimens already exceed this solubilization capacity, increasing dosing frequency delivers more drug but does not increase the amount actually absorbed. Clinically, this suggests that for children not achieving target exposure on TID, switching to QID is unlikely to provide meaningful benefit, and alternative formulations should be considered instead.

This study has several limitations. First, although the pediatric model was developed and validated using clinical data from children with hematologic malignancies and demonstrated a good predictive performance, the underlying physiological parameters were derived from healthy children. Disease states may alter gastrointestinal function or drug metabolism, potentially introducing bias when extrapolating to other pediatric populations. Second, the simulated 90% CIs were wider than observed variability, suggesting that our model overestimates the variability in absorption-related parameters. This reduces individual-level precision but preserves population-level accuracy. Third, the meal composition was simulated based on the standardized fat and calorie content. However, real-world dietary intake varies considerably among individuals, warranting caution when interpreting fed-state simulation results. This limitation is particularly relevant for children aged < 2 years, as infant feeding patterns (eg, milk-based, frequent small-volume feedings) differ substantially from adult meals. Extrapolation to this youngest age group should be approached with caution. In addition, as the current model is specifically developed to characterize the food effect, it does not yet include dedicated simulation workflows for DDIs with commonly co-administered medications such as PPIs and chemotherapeutic agents.

Conclusion

This study provides a quantitative pharmacological understanding of the combined effects of age, meal composition, dose, and dosing frequency on the PK of posaconazole oral suspension in pediatric patients. Differentiated dosing and dietary strategies should be adopted according to the age group. Further prospective clinical validation is still required, and DDI simulation modules will be incorporated in subsequent model optimization to support its real-world dosing decision application. Furthermore, given the marked inter-individual PK variability and saturated absorption characteristic of posaconazole, routine TDM is strongly recommended in clinical practice to ensure optimal efficacy.

Data Sharing Statement

The original contributions presented in this study are included in the article, and further inquiries can be directed to the corresponding author Juan Wu.

Ethical Approval

This study was conducted in accordance with the principles of the Declaration of Helsinki and approved by the Ethics Committee of Shanghai Children's Medical Center (SCMCIRB-K2022029-1). All guardians of the patients provided written informed consent prior to enrollment. This study was registered with the [chictr.gov.cn](https://www.chictr.gov.cn) identifier ChiCTR2200059843 (registered on 2022-05-12).

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Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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Disclosure

The authors have no conflicts of interest to declare regarding the content of this article.

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